Technical Reproducibility Validation Report

STRaitRazor

Version v3
STR / SNP panel

Verification date	2026-11-06
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Result	**Runs as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/straitrazor
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs as documented**
REACHED	Runs + Plausible output, 6/6 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it runs.

A third party installed the tool with the steps this commit states and ran the command it writes, and the output file that command names appeared. This says nothing about genotypes or about the forensic panel.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

4. What the Author Documents as a Known Issue

Documented by the author in the official repository. Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

known issues (Official README line 114)

> There's been one computer architecture (windows 7 + Xeon processor) that's caused some issues with the str8rzr.exe executable. In this case, str8rzr would occassionally crash, and I had to recompile it for that computer (and since then it's been fine). Please let me know if you experience problems-- especially crashes-- it'll let me further diagnose the exact problems therein. str8rzr is written in C/C++ with multithreading support using the pthreads library. This new release couples a new search strategy (see algorithm) coupled with a complete redesign of the code-base used to identify short …

5. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRaitRazor v3 · commit b618e93
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( str8rzr \
-c ForenSeqv1.27.config \
/data/in/sample.fastq > allsequences.txt ); rc=$?; find . \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

6. Reproducibility Metadata

Tool	STRaitRazor
Version	v3
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-11-06 16:16:46 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42250554834

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

7. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

8. Output Content Evidence

Output file	allsequences.txt
Format	TSV
Sequence records	816
Distinct STR loci	63
Total reads	3,967
Max depth (single locus)	132

9. Verification Data

This tool ships no demo or test data of its own. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D17S1301 · D18S51 · D19S433 · D1S1656 · D20S482 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D4S2408 · D5S818 · D6S1043 · D7S820 · D8S1179 · D9S1122 · DXS10074 · DXS10103 · DXS10135 · DXS7132 · DXS7423 · DXS8378 · DYF387S1 · DYS19 · DYS385 · DYS389I · DYS389II · DYS390 · DYS391 · DYS392 · DYS437 · DYS438 · DYS439 · DYS448 · DYS460 · DYS461 · DYS481 · DYS505 · DYS522 · DYS533 · DYS549 · DYS570 · DYS576 · DYS612 · DYS635 · DYS643 · FGA · HPRTB · N29insA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · mh16-MC1RB · mh16-MC1RC · vWA

10. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

11. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

12. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's Makefile (make), at the pinned commit.
– Input: the README documents FASTQ (line 19); this run used STRhub's illumina-str-fastq reference data.
– 'str8rzr' config placeholder resolved to ForenSeqv1.27.config (matches the dataset's ForenSeq kit).
– 1 input placeholder(s) in the README command replaced with /data/in/sample.fastq.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.

13. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

14. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

15. Conclusion

- **Runs end-to-end**

  STRaitRazor v3 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

- **Produces plausible output**

  The tool generated a structurally valid TSV output file with genotype calls across 63 target forensic STR loci, with a maximum read depth of 132 at DYS392.

- **Verdict: Runs**

  The tool installed and its run produced its documented output.

- **No bundled demo data**

  STRaitRazor ships no test or demo data of its own. STRhub Verified supplied NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth (1/3)

STRaitRazor v3 · verified 2026-11-06 · showing the 120 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH	LOCUS	READ DEPTH
DYS392	170	**D22S1045**	44
DYS438	117	**DXS8378**	44
D6S1043	111	**DXS7423**	43
TH01	106	**rs1109037**	43
DYS389I	97	**DYS437**	40
DYS576	95	**mh16-MC1RC**	40
DYS505	93	**DYS448**	39
D20S482	84	**D10S1248**	38
FGA	81	**rs459920**	38
D3S1358	69	**D7S820**	37
D2S441	66	**DYS643**	37
DYS612	63	**HPRTB**	36
D16S539	61	**rs4364205**	36
D19S433	59	**D8S1179**	34
DXS10074	58	**DXS10135**	34
D4S2408	52	**DYF387S1**	34
DXS7132	51	**DYS390**	34
D18S51	48	**D12S391**	31
rs200354	47	**DYS391**	31
DYS385	45	**D21S11**	30

Appendix: Detected Markers with Read Depth (2/3)

STRaitRazor v3 · verified 2026-11-06 · showing the 120 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH	LOCUS	READ DEPTH
rs1919550	29	**rs4411548**	20
D13S317	28	**rs993934**	20
rs174570	28	**rs1876482**	19
rs192655	28	**rs2024566**	19
D17S1301	26	**rs735480**	19
CSF1PO	24	**rs8037429**	19
rs1454361	24	**PentaE**	18
rs8078417	24	**TPOX**	18
DYS389II	23	**rs2040411**	18
DYS439	23	**rs798443**	18
DYS570	23	**rs2166624**	17
rs1382387	23	**rs737681**	17
rs722098	23	**rs9522149**	17
PentaD	22	**D2S1338**	16
rs2814778	22	**DYS522**	16
rs876724	22	**rs1335873**	16
rs2402130	21	**rs1800407**	16
rs4918664	21	**rs2196051**	16
rs917118	21	**rs560681**	16
DYS481	20	**rs671**	16

Appendix: Detected Markers with Read Depth (3/3)

STRaitRazor v3 · verified 2026-11-06 · showing the 120 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH	LOCUS	READ DEPTH
rs2042762	15	**rs7657799**	11
rs251934	15	**D1S1656**	10
rs3827760	15	**D5S818**	10
rs1490413	14	**DYS549**	10
rs4530059	14	**mh16-MC1RB**	10
rs7722456	14	**rs1800414**	10
rs7997709	14	**rs1805009**	10
rs1005533	13	**rs1821380**	10
rs722290	13	**rs2111980**	10
rs735155	13	**rs260690**	10
rs11652805	12	**Y-GATA-H4**	9
rs1979255	12	**N29insA**	9
rs2593595	12	**rs1058083**	9
rs2831700	12	**rs1229984**	9
rs4959270	12	**rs2399332**	9
rs740910	12	**rs3780962**	9
rs9905977	12	**rs6754311**	9
rs12439433	11	**rs7251928**	9
rs1871534	11	**rs964681**	9
rs430046	11	**rs1042602**	8